Gagan Jain

AI / LLM Engineer

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PROFESSIONAL SUMMARY

AI/LLM Engineer with hands-on experience building and deploying production-grade GenAI systems, agentic AI frameworks, and LLM-powered applications. Core strengths include RAG pipelines, multi-agent orchestration, LLM fine-tuning (QLoRA/LoRA), prompt and context engineering, MCP tooling, and evaluation/observability. Strong Python foundation with end-to-end deployment experience using Docker, CI/CD, and cloud platforms.

TECHNICAL SKILLS

AI / LLM: LLM Integration, RAG Pipelines, Agentic AI Frameworks, Prompt Engineering, Context Engineering, Multi-Agent Systems, Tool-Using Agents, Planning Agents, Fine-tuning (QLoRA/LoRA), MCP / Tool Protocols, LangChain / LlamaIndex, Vector Databases, Evaluation & Observability, AI Governance, Local LLM Deployment

Languages: Python, TypeScript, JavaScript, Rust, PHP, C, C++, HTML/CSS

Backend & APIs: Flask, FastAPI, Node.js, Express, Laravel, Axum (Rust), REST APIs, WebSockets, FastMCP

Databases: MySQL, MongoDB, Redis, SQLite, Supabase

DevOps & Tooling: Docker, GitHub Actions, Nginx, Gunicorn, Prometheus, Linux, CI/CD Pipelines

Frontend: React, React Native, Next.js, Tailwind CSS, Bootstrap, Monaco Editor

PROJECTS

NexusAOS AI/AGENTIC

Python, FastMCP, QLoRA, Unsloth, Multi-Agent, AsyncIO, Rust, Zig

- Built multi-agent swarm executor with collision detection, namespace isolation, quorum voting, and atomic fission in Python + asyncio
- Integrated local LLM inference pipeline with Phi-4-Mini QLoRA adapters, AB/AP balance enforcement, and constitution-guided routing
- Exposed 30+ MCP tools via FastMCP for metabolic, planning, immune, and physical substrate control
- Designed adapter-routed inference system with curriculum learning, evaluation harness, and model benchmarking

nexus-kernel RUST

Rust, Tokio, OpenAI, Anthropic, SQLite, Ratatui, Iced, SSH, Event Sourcing

- Built Rust 2024 microkernel with 12 workspace crates, 981 passing tests, 0 clippy warnings, and full CI/CD
- Implemented event-sourced append-only audit trail, policy engine with trust tiers, and provider-swappable model interface
- Integrated OpenAI-compatible and Anthropic streaming with real-time token streaming into TUI/GUI
- Delivered native terminal emulation (PTY + VT100 + Zig parser), SSH multiplexing, and multi-interface CLI/TUI/GUI/RPC

SeshaOS AI/AGENTIC

Rust, LiteLLM, Gemma, Qwen, Local LLMs, Governance, Event Sourcing, Ubuntu

- Architected specialist model stack: Gemma 4 12B (Planner), Qwen3-Coder 30B (Implementation), Qwen3.5 9B (Vision)
- Designed kernel-centric governance where models propose actions and the kernel validates/records every state change
- Implemented LiteLLM-compatible proxy support for NVIDIA NIM and other model providers
- Maintained event-sourced architecture with reversible, permissioned actions and offline-first execution

Vyākṛti COMPILER

Rust, React, TypeScript, Monaco Editor, Zustand, Tailwind CSS, Axum

- Complete compiler pipeline: lexer → parser → type checker → bytecode compiler — all built from scratch in Rust
- Browser-based IDE with React, Monaco Editor, syntax highlighting, autocomplete, and diagnostics
- Rust (axum) backend with compile, REPL, LSP, and file management endpoints via REST + WebSocket
- 123 tests covering the full pipeline, including a self-hosting corpus

EXPERIENCE

Industry 5.0 Industrial Automation Trainee

Aug 2025 – Nov 2025

CodenPlay Robotics

- Programmed PLC control logic for automated production workflows using CODESYS structured text and ladder logic
- Designed HMI monitoring dashboards and multi-device communication networks using Node-RED
- Validated automation sequences against Factory I/O digital-twin simulations prior to hardware deployment

EDUCATION

Vellore Institute of Technology, Vellore

2026

B.Tech in Computer Science and Engineering

CGPA: 7.7 / 10.0 | Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Web Technologies, Artificial Intelligence, Machine Learning